

Laboratório 4

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Questao 1

```
library(readxl)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.2      v tibble     3.2.1
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.0.4
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

Questao 2

```
fname <- "WDIEXCEL.xlsx"
file.exists(fname)
```

```
[1] TRUE
```

Questao 3

```
countries <- read_excel(fname, sheet = "Country")
countries
```

```
# A tibble: 264 x 31
  `Country Code` `Short Name` `Table Name` `Long Name` `2-alpha code`
  <chr>          <chr>          <chr>      <chr>      <chr>
1 ABW           Aruba           Aruba      Aruba      AW
2 AFE           Africa Eastern and So~ Africa East~ Africa Eas~ ZH
3 AFG           Afghanistan      Afghanistan Islamic St~ AF
4 AFW           Africa Western and Ce~ Africa West~ Africa Wes~ ZI
5 AGO           Angola           Angola      People's R~ AO
6 ALB           Albania          Albania      Republic o~ AL
7 AND           Andorra          Andorra      Principali~ AD
8 ARB           Arab World       Arab World   Arab World  1A
9 ARE           United Arab Emirates United Arab~ United Ara~ AE
10 ARG          Argentina       Argentina   Argentine ~ AR
# i 254 more rows
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, ...
```

Questao 4

```
# Brasil + 4 outros países escolhidos
países_escolhidos <- c("BRA", "USA", "CHN", "IND", "ZAF")

countries <- countries |>
  filter(`Country Code` %in% países_escolhidos)

countries
```

```
# A tibble: 5 x 31
  `Country Code` `Short Name` `Table Name` `Long Name` `2-alpha code`
  <chr>          <chr>          <chr>      <chr>      <chr>
1 BRA           Brazil          Brazil      Federative Republic~ BR
2 CHN           China          China      People's Republic o~ CN
3 IND           India          India      Republic of India    IN
4 USA           United States United States United States of Am~ US
5 ZAF           South Africa  South Africa Republic of South A~ ZA
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
```

```
# `Lending category` <chr>, `Other groups` <chr>,
# `System of National Accounts` <chr>, `Alternative conversion factor` <lgl>,
# `PPP survey year` <lgl>, `Balance of Payments Manual in use` <chr>, ...
```

Questao 5

```
WDI <- read_excel(fname, sheet = "Data")
WDI
```

```
# A tibble: 396,970 x 70
  `Country Name` `Country Code` `Indicator Name` `Indicator Code` `1960` `1961`
    <chr>         <chr>         <chr>         <chr>         <dbl> <dbl>
1 Africa Easter~ AFE           Access to clean~ EG.CFT.ACCS.ZS      NA      NA
2 Africa Easter~ AFE           Access to clean~ EG.CFT.ACCS.RU.~    NA      NA
3 Africa Easter~ AFE           Access to clean~ EG.CFT.ACCS.UR.~    NA      NA
4 Africa Easter~ AFE           Access to elect~ EG.ELC.ACCS.ZS      NA      NA
5 Africa Easter~ AFE           Access to elect~ EG.ELC.ACCS.RU.~    NA      NA
6 Africa Easter~ AFE           Access to elect~ EG.ELC.ACCS.UR.~    NA      NA
7 Africa Easter~ AFE           Account ownersh~ FX.OWN.TOTL.ZS      NA      NA
8 Africa Easter~ AFE           Account ownersh~ FX.OWN.TOTL.FE.~    NA      NA
9 Africa Easter~ AFE           Account ownersh~ FX.OWN.TOTL.MA.~    NA      NA
10 Africa Easter~ AFE           Account ownersh~ FX.OWN.TOTL.OL.~    NA      NA
# i 396,960 more rows
# i 64 more variables: `1962` <dbl>, `1963` <dbl>, `1964` <dbl>, `1965` <dbl>,
# `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>, `1970` <dbl>,
# `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>, `1975` <dbl>,
# `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>, `1980` <dbl>,
# `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>, `1985` <dbl>,
# `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>, `1990` <dbl>, ...
```

Questao 6

```
WDI <- WDI |>
  select(`Country Code`, `Indicator Code`, `1960`:`2018`)
WDI
```

```
# A tibble: 396,970 x 61
  `Country Code` `Indicator Code` `1960` `1961` `1962` `1963` `1964` `1965`
    <chr>         <chr>         <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
```

```

1 AFE          EG.CFT.ACCS.ZS          NA      NA      NA      NA      NA      NA
2 AFE          EG.CFT.ACCS.RU.ZS        NA      NA      NA      NA      NA      NA
3 AFE          EG.CFT.ACCS.UR.ZS        NA      NA      NA      NA      NA      NA
4 AFE          EG.ELC.ACCS.ZS           NA      NA      NA      NA      NA      NA
5 AFE          EG.ELC.ACCS.RU.ZS        NA      NA      NA      NA      NA      NA
6 AFE          EG.ELC.ACCS.UR.ZS        NA      NA      NA      NA      NA      NA
7 AFE          FX.OWN.TOTL.ZS           NA      NA      NA      NA      NA      NA
8 AFE          FX.OWN.TOTL.FE.ZS        NA      NA      NA      NA      NA      NA
9 AFE          FX.OWN.TOTL.MA.ZS        NA      NA      NA      NA      NA      NA
10 AFE         FX.OWN.TOTL.OL.ZS        NA      NA      NA      NA      NA      NA
# i 396,960 more rows
# i 53 more variables: `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>,
#   `1970` <dbl>, `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>,
#   `1975` <dbl>, `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>,
#   `1980` <dbl>, `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>,
#   `1985` <dbl>, `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>,
#   `1990` <dbl>, `1991` <dbl>, `1992` <dbl>, `1993` <dbl>, `1994` <dbl>, ...

```

Questao 7

Os dados NÃO estão em formato tidy, pois cada ano está espalhado em uma coluna diferente, quando deveria existir uma única coluna “YEAR” e uma coluna de valor.

```

WDI <- WDI |>
  pivot_longer(
    cols      = `1960`:`2018`,
    names_to  = "YEAR",
    values_to = "VALUE"
  )

WDI

```

```

# A tibble: 23,421,230 x 4
  `Country Code` `Indicator Code` YEAR  VALUE
  <chr>          <chr>          <chr> <dbl>
1 AFE          EG.CFT.ACCS.ZS    1960    NA
2 AFE          EG.CFT.ACCS.ZS    1961    NA
3 AFE          EG.CFT.ACCS.ZS    1962    NA
4 AFE          EG.CFT.ACCS.ZS    1963    NA
5 AFE          EG.CFT.ACCS.ZS    1964    NA
6 AFE          EG.CFT.ACCS.ZS    1965    NA
7 AFE          EG.CFT.ACCS.ZS    1966    NA
8 AFE          EG.CFT.ACCS.ZS    1967    NA

```

```

9 AFE          EG.CFT.ACCS.ZS    1968    NA
10 AFE         EG.CFT.ACCS.ZS    1969    NA
# i 23,421,220 more rows

```

Questao 8

```
class(WDI$YEAR)
```

```
[1] "character"
```

```

# year não é inteiro (veio como character do pivot_longer), então:
WDI <- WDI |>
  mutate(YEAR = as.integer(YEAR))

class(WDI$YEAR)

```

```
[1] "integer"
```

Questao 9

```

WDI <- WDI |>
  filter(`Country Code` %in% countries$`Country Code`) |>
  filter(`Indicator Code` == "SP.DYN.CBRT.IN") |>
  rename(SP.DYN.CBRT.IN = VALUE) |>
  inner_join(countries |> select(`Country Code`, `Short Name`),
    by = "Country Code") |>
  select(`Country Code`, `Short Name`, YEAR, SP.DYN.CBRT.IN)

```

```
WDI
```

```

# A tibble: 295 x 4
  `Country Code` `Short Name`  YEAR SP.DYN.CBRT.IN
  <chr>          <chr>      <int>      <dbl>
1 BRA           Brazil      1960       43.8
2 BRA           Brazil      1961       43.3
3 BRA           Brazil      1962       42.7
4 BRA           Brazil      1963       42.0
5 BRA           Brazil      1964       41.0
6 BRA           Brazil      1965       39.8

```

```

7 BRA          Brazil      1966      38.6
8 BRA          Brazil      1967      37.4
9 BRA          Brazil      1968      36.3
10 BRA         Brazil      1969      35.7
# i 285 more rows

```

Questao 10

```

WDI <- WDI |>
  arrange(`Country Code`, YEAR)

WDI

```

```

# A tibble: 295 x 4
  `Country Code` `Short Name` YEAR SP.DYN.CBRT.IN
  <chr>          <chr>    <int>    <dbl>
1 BRA          Brazil    1960     43.8
2 BRA          Brazil    1961     43.3
3 BRA          Brazil    1962     42.7
4 BRA          Brazil    1963     42.0
5 BRA          Brazil    1964     41.0
6 BRA          Brazil    1965     39.8
7 BRA          Brazil    1966     38.6
8 BRA          Brazil    1967     37.4
9 BRA          Brazil    1968     36.3
10 BRA         Brazil    1969     35.7
# i 285 more rows

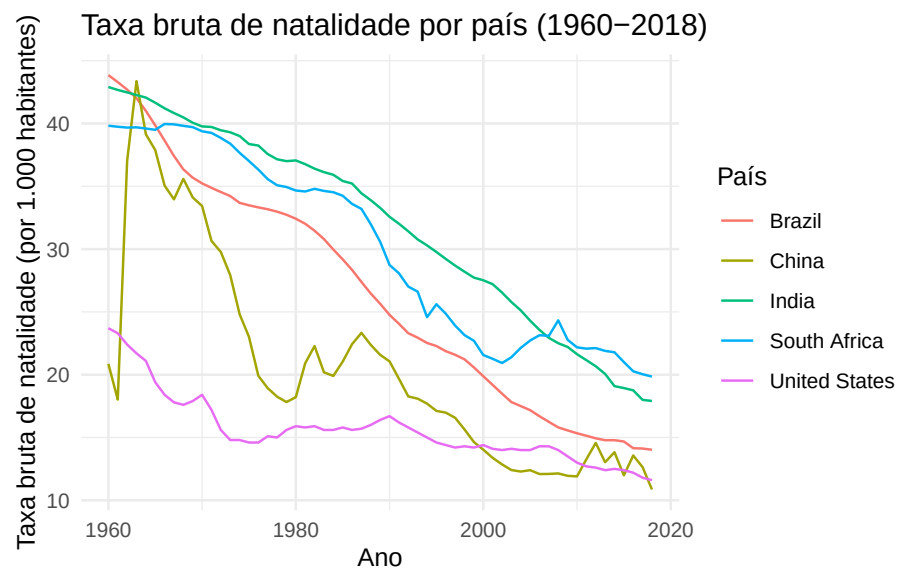
```

Questao 11

```

ggplot(WDI, aes(x = YEAR, y = SP.DYN.CBRT.IN, color = `Short Name`)) +
  geom_line() +
  labs(
    title = "Taxa bruta de natalidade por país (1960-2018)",
    x = "Ano",
    y = "Taxa bruta de natalidade (por 1.000 habitantes)",
    color = "País"
  ) +
  theme_minimal()

```



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-05 22:24:52 -03"
```